

SRAVYA ISUKAPATLA

sravyaisukapatla@gmail.com | sravya.com | sravya.linkedin | sravya.github

PROFILE SUMMARY

Computer Science student with experience in full-stack development, IoT systems, and real-time data processing. Skilled in Python, JavaScript, Dart, and backend development with hands-on experience building scalable applications and sensor-based monitoring systems. Experienced with Flutter, Node.js, MQTT, REST APIs, and containerized applications. Interested in backend engineering, cloud computing, and distributed systems.

SKILLS

Programming: Python, JavaScript, Dart, C, Java, HTML, CSS
Frameworks & Tools: Flutter, Node.js, Flask, React, Git
Databases: MongoDB, Firebase, Supabase
Cloud Technologies: AWS (EC2, S3, IAM, VPC – learning)
DevOps: Docker, Git, Linux CLI
Networking: TCP/IP, DNS, HTTP, REST APIs
Technologies: MQTT, Real-time data systems, API integration

EDUCATION

PONDICHERRY UNIVERSITY – BSc Computer Science (Honours)	2023-2027
ASHRAM PUBLIC SCHOOL – Higher Secondary (CBSE), 83%	2023
ST. JOSEPH CONVENT SCHOOL – ICSE, 10th Grade 80%	2021

EXPERIENCE

INTERN - MOBILE APP DEVELOPMENT ↗, ONGC – Dehradun	May-July 2024
- Developed a cross-platform mobile application using Flutter and Dart with Firebase backend integration. - Built responsive interfaces providing real-time access to event schedules, notifications, and updates. - Implemented backend data synchronization for dynamic event management and user interaction.	
INTERN - UNIVERSITY OF HYDERABAD AirSense 5G ↗	2024-2025
- Developed an IoT-based air-quality monitoring application integrating 5 MQTT sensors with real-time data streaming. - Built a Flask backend integrating a local Small Language Model (Phi-2) via LM Studio for intelligent air-quality insights. - Implemented a machine learning prediction pipeline using Linear Regression for environmental metrics. - Designed a real-time data pipeline converting MQTT sensor streams into structured datasets with dashboard updates every 30 seconds.	
RESEARCH INTERN - IIT Madras Energy Consortium ↗ Digital Twin Based Smart Grid Automation System (Ongoing)	May-July 2026
- Designing a Digital Twin framework for the IEEE-33 bus distribution network to simulate and monitor power distribution systems. - Exploring OpenDSS integration for power flow analysis, voltage monitoring, and fault simulation. - Planning a Docker-based microservice architecture for voltage monitoring, fault detection, switching control, and service restoration. - Researching the use of Retrieval-Augmented Generation (RAG) to enable natural language interaction with grid data and operational information. - Investigating cloud-native approaches for scalable smart grid automation and decision support.	
INTERN - SeraphGuard Labs ↗	March 2026

- Developing cross-platform mobile interfaces using Flutter for the Guardian AI parental control platform.
- Implementing real-time monitoring dashboards for digital activity insights and safety alerts.
- Integrating backend APIs and collaborating with AI teams to support intelligent monitoring features.

PROJECTS

Cloud Deployment Project

- Containerized a web application using Docker for portable deployment.
- Implemented REST API backend and static frontend architecture.
- Designed a cloud-ready application structure suitable for deployment on AWS infrastructure.

Hackathon – CareConnect ↗

- Secured 1st place at the Sustainable Healthcare Hackathon.
- Built a healthcare platform using React, Node.js, and Flutter with role-based dashboards.
- Integrated MQTT-based IoT monitoring for real-time patient vitals.

ShopLocal – E-commerce Platform ↗

- Developed a full-stack e-commerce platform using React and Vite with a Supabase backend.
- Implemented authentication and server security using bcrypt and Helmet.
- Optimized frontend performance and added interactive animations using GSAP.

CERTIFICATIONS & WORKSHOPS

NPTEL Ethical Hacking Certification ↗

- Completed coursework covering vulnerability assessment and network security fundamentals.

AI/ML for GeoData Analysis (ISRO) ↗

- Training in geo-data processing and machine learning applications for satellite imagery.

Drone Security Workshop ↗

- Hands-on experience securing RF/MAVLink communication protocols using Kali Linux.

TECHNICAL INTERESTS

Cloud Computing

Backend Systems

Distributed Systems

DevOps

IoT Platforms